

作业二：基于 QEMU 的 OpenWrt 仿真实验报告

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一、实验目的

本实验旨在基于 QEMU 搭建 OpenWrt 仿真环境，在虚拟化条件下运行适用于 ARM 架构的 OpenWrt 镜像，验证系统能否正常启动与运行；在此基础上，通过安装并访问 LuCI 管理界面，验证 OpenWrt 的 Web 服务功能是否正常；同时，利用 SSH 方式远程连接 OpenWrt 系统，验证其远程管理服务的可用性，从而加深对嵌入式 Linux 系统仿真、网络服务配置及远程管理方法的理解。

二、实验环境

- 计算机型号：Apple MacBook Air
 - 处理器型号：Apple M4
 - 内存容量：24 GB
 - 宿主机操作系统：macOS Sequoia 15.6.1
 - 虚拟机操作系统：Kali Linux
 - 虚拟化软件：QEMU
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三、实验步骤

1. 镜像下载与准备

新建一个专门存放 OpenWrt 镜像和启动文件的目录，避免后面文件混乱：

```
(parallels@kali-linux-2024-2)-[~]  
└─$ mkdir ~/openwrt  
  
(parallels@kali-linux-2024-2)-[~]  
└─$ cd ~/openwrt
```

下载 OpenWrt 镜像并解压，接着下载引导文件 u-boot.bin：

```
wget  
https://downloads.openwrt.org/releases/23.05.2/targets/armv8/openwrt-  
23.05.2-armv8-generic-squashfs-combined.img.gz  
gzip -d openwrt-23.05.2-armv8-generic-squashfs-combined.img.gz  
wget https://mirror-  
03.infra.openwrt.org/releases/23.05.2/targets/armv8/u-boot-  
qemu_armv8/u-boot.bin
```

```
parallels@kali-linux-2024-2: ~/openwrt
File Actions Edit View Help
Trash
(parallels@kali-linux-2024-2)-[~/openwrt]
$ gzip -d openwrt-23.05.2-armv8-generic-squashfs-combined.img.gz

gzip: openwrt-23.05.2-armv8-generic-squashfs-combined.img.gz: decompression OK, trailing garbage ignored

(parallels@kali-linux-2024-2)-[~/openwrt]
$ wget https://mirror-03.infra.openwrt.org/releases/23.05.2/targets/armv8/u-boot-qemu_armv8/u-boot.bin
--2026-03-17 19:03:22-- https://mirror-03.infra.openwrt.org/releases/23.05.2/targets/armv8/u-boot-qemu_armv8/u-boot.bin
Resolving mirror-03.infra.openwrt.org (mirror-03.infra.openwrt.org) ... 198.18.3.53
Connecting to mirror-03.infra.openwrt.org (mirror-03.infra.openwrt.org)|198.18.3.53|:443 ..
. connected.
HTTP request sent, awaiting response... 200 OK
Length: 998192 (975K) [application/octet-stream]
Saving to: 'u-boot.bin'

u-boot.bin          100%[=====>] 974.80K  1.19MB/s   in 0.8s
2026-03-17 19:03:24 (1.19 MB/s) - 'u-boot.bin' saved [998192/998192]
```

检查文件是否下载成功:

```
(parallels@kali-linux-2024-2)-[~/openwrt]
$ ls -lh
```

```
(parallels@kali-linux-2024-2)-[~/openwrt]
$ ls -lh
total 234M
-rw-rw-r-- 1 parallels parallels 233M Nov 15 2023 openwrt-23.05.2-armv8-generic-squashfs-combined.img
-rw-rw-r-- 1 parallels parallels 975K Nov 15 2023 u-boot.bin
```

✅ 镜像准备成功

2. QEMU 环境配置

先检查 QEMU 是否安装成功:

```
qemu-system-aarch64 --version
```

```
(parallels@kali-linux-2024-2)-[~/openwrt]
$ qemu-system-aarch64 --version
QEMU emulator version 10.2.0 (Debian 1:10.2.0+ds-2)
Copyright (c) 2003-2025 Fabrice Bellard and the QEMU Project developers
```

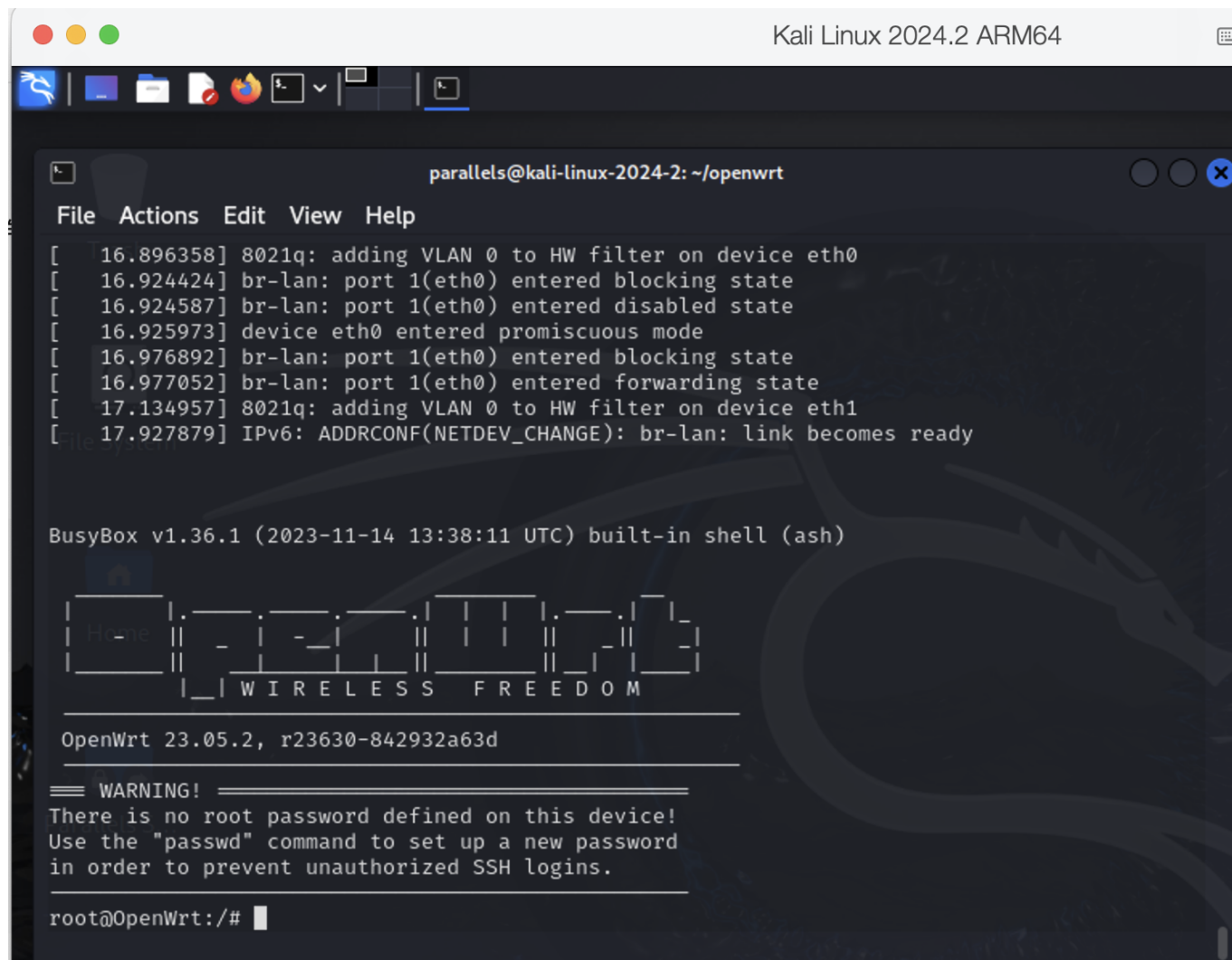
终端能显示版本号，可以继续。

启动 OpenWrt 虚拟机:

```
qemu-system-aarch64 -cpu cortex-a72 -m 1024 -M virt,highmem=off -nographic \
-bios u-boot.bin \
-drive file=openwrt-23.05.2-armv8-generic-squashfs-combined.img,format=raw,if=virtio \
-device virtio-net,netdev=net0 -netdev user,id=net0,net=192.168.1.0/24,hostfwd=tcp:127.0.0.1:1122-192.168.1.1:22,hostfwd=tcp:127.0.0.1:8080-192.168.1.1:80 \
-device virtio-net,netdev=net1 -netdev user,id=net1,net=192.168.2.0/24
```

上述命令参数简要说明：

- `qemu-system-aarch64`：启动 AArch64（ARMv8 64 位）虚拟机。
- `-cpu cortex-a72`：指定虚拟 CPU 型号为 Cortex-A72。
- `-m 1024`：分配 1024 MB 内存。
- `-M virt,highmem=off`：使用通用 `virt` 机器类型；关闭 `highmem`（减少地址空间/设备映射复杂度，提升兼容性）。
- `-nographic`：不启用图形界面，串口输出直接显示在当前终端。
- `-bios u-boot.bin`：指定引导固件（这里用 U-Boot）用于启动系统。
- `-drive file=...,format=raw,if=virtio`：把 OpenWrt 镜像作为磁盘挂载；`raw` 为镜像格式；`virtio` 为高性能虚拟磁盘接口。
- `-netdev user,id=net0,net=192.168.1.0/24`：创建一个用户态网络后端（NAT），并指定虚拟网段。
- `-device virtio-net,netdev=net0`：添加一块 virtio 网卡并连接到 `net0`。
- `hostfwd=tcp:127.0.0.1:1122-192.168.1.1:22`：把宿主机 `127.0.0.1:1122` 转发到虚拟机 `192.168.1.1:22`（SSH）。
- `hostfwd=tcp:127.0.0.1:8080-192.168.1.1:80`：把宿主机 `127.0.0.1:8080` 转发到虚拟机 `192.168.1.1:80`（Web/LuCI）。
- 第二组 `-netdev user,id=net1,net=192.168.2.0/24 + -device virtio-net,netdev=net1`：再添加一套用户态网络与第二块网卡（第二个虚拟网段）。



The screenshot shows a terminal window titled "parallels@kali-linux-2024-2: ~/openwrt". The terminal displays the following logs:

```
[ 16.896358] 8021q: adding VLAN 0 to HW filter on device eth0
[ 16.924424] br-lan: port 1(eth0) entered blocking state
[ 16.924587] br-lan: port 1(eth0) entered disabled state
[ 16.925973] device eth0 entered promiscuous mode
[ 16.976892] br-lan: port 1(eth0) entered blocking state
[ 16.977052] br-lan: port 1(eth0) entered forwarding state
[ 17.134957] 8021q: adding VLAN 0 to HW filter on device eth1
[ 17.927879] IPv6: ADDRCONF(NETDEV_CHANGE): br-lan: link becomes ready
```

Below the logs, the terminal shows the BusyBox v1.36.1 (2023-11-14 13:38:11 UTC) built-in shell (ash) prompt. A banner for OpenWrt 23.05.2, r23630-842932a63d is displayed, along with a warning message:

```
===== WARNING! =====
There is no root password defined on this device!
Use the "passwd" command to set up a new password
in order to prevent unauthorized SSH logins.
=====
```

The terminal prompt is now root@OpenWrt:/#.

✅ OpenWrt 已经启动成功

3. 启动网页界面

安装 LuCI 网页管理界面:

```
opkg update
opkg install luci
```

```
parallels@kali-linux-2024-2: ~/openwrt
File Actions Edit View Help
Downloading https://downloads.openwrt.org/releases/23.05.2/packages/aarch64_generic/luci/P
ackages.sig
Signature check passed.
Downloading https://downloads.openwrt.org/releases/23.05.2/packages/aarch64_generic/packag
es/Packages.gz
Updated list of available packages in /var/opkg-lists/openwrt_packages
Downloading https://downloads.openwrt.org/releases/23.05.2/packages/aarch64_generic/packag
es/Packages.sig
Signature check passed.
Downloading https://downloads.openwrt.org/releases/23.05.2/packages/aarch64_generic/routin
g/Packages.gz
Updated list of available packages in /var/opkg-lists/openwrt_routing
Downloading https://downloads.openwrt.org/releases/23.05.2/packages/aarch64_generic/routin
g/Packages.sig
Signature check passed.
Downloading https://downloads.openwrt.org/releases/23.05.2/packages/aarch64_generic/teleph
ony/Packages.gz
Updated list of available packages in /var/opkg-lists/openwrt_telephony
Downloading https://downloads.openwrt.org/releases/23.05.2/packages/aarch64_generic/teleph
ony/Packages.sig
Signature check passed.
root@OpenWrt:/# opkg install luci
Upgrading luci on root from git-23.051.66410-a505bb1 to git-25.222.75657-7ce34fe ...
Downloading https://downloads.openwrt.org/releases/23.05.2/packages/aarch64_generic/luci/l
uci_git-25.222.75657-7ce34fe_all.ipk
Configuring luci.
root@OpenWrt:/#
```

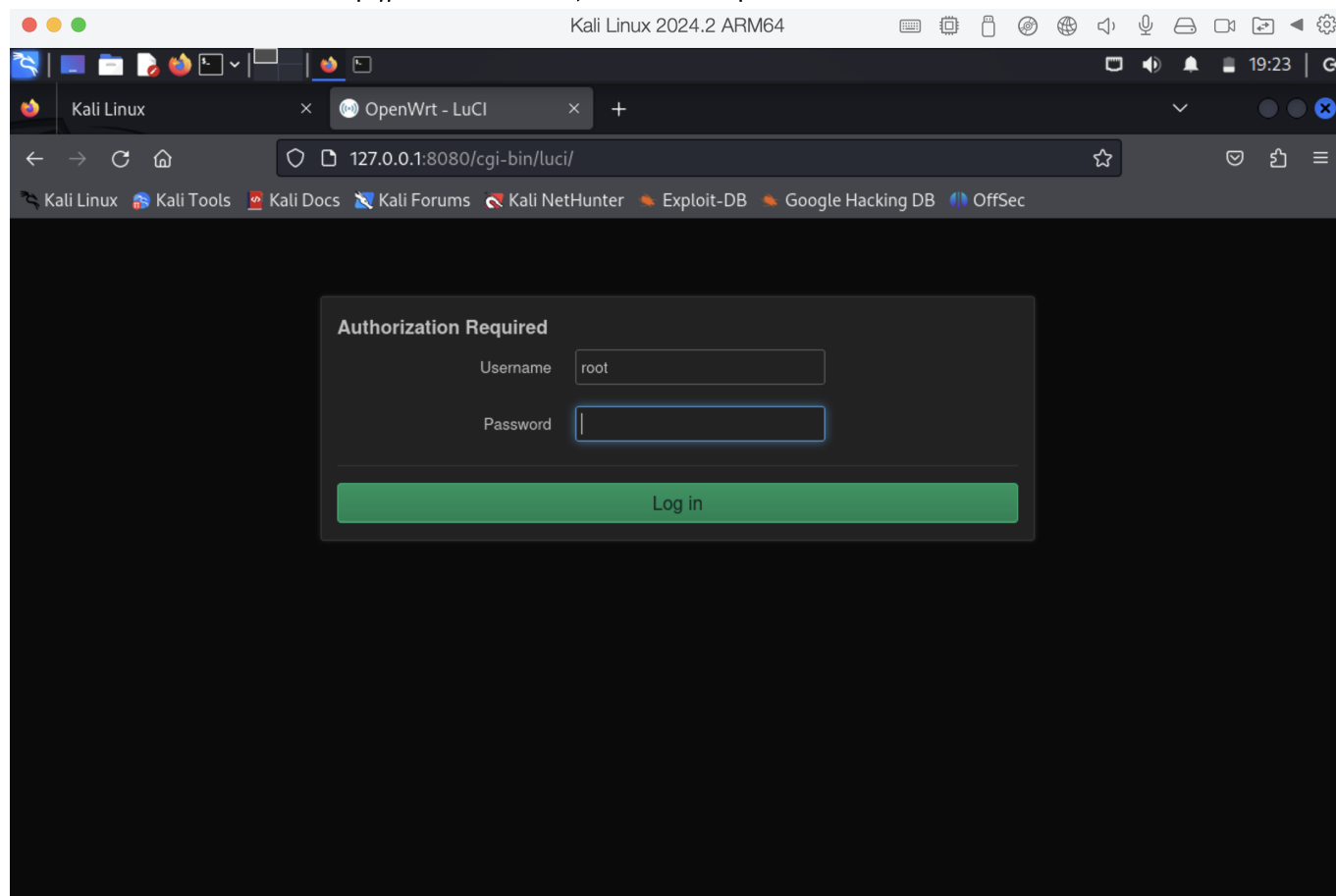
检查网页服务是否已经起来:

```
netstat -tupan
```

可以看到 **uhttpd** 已经启动，80 端口正常监听，网页界面可以访问：

```
root@OpenWrt:/# netstat -tupan
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.1:53            0.0.0.0:*               LISTEN      2466/dnsmasq
tcp        0      0 192.168.1.1:53          0.0.0.0:*               LISTEN      2466/dnsmasq
tcp        0      0 0.0.0.0:443             0.0.0.0:*               LISTEN      1173/uhttpd
tcp        0      0 0.0.0.0:80              0.0.0.0:*               LISTEN      1173/uhttpd
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN      1683/dropbear
tcp        0      0 192.168.2.15:53         0.0.0.0:*               LISTEN      2466/dnsmasq
tcp        0      0 192.168.2.15:50598      198.18.3.52:443        TIME_WAIT   -
tcp        0      0 :::443                  :::*                    LISTEN      1173/uhttpd
tcp        0      0 fe80::5054:ff:fe12:3456:53 :::*                    LISTEN      2466/dnsmasq
tcp        0      0 :::80                   :::*                    LISTEN      1173/uhttpd
tcp        0      0 fe80::5054:ff:fe12:3457:53 :::*                    LISTEN      2466/dnsmasq
tcp        0      0 fd76:3a89:d79a::1:53    :::*                    LISTEN      2466/dnsmasq
tcp        0      0 :::22                   :::*                    LISTEN      1683/dropbear
tcp        0      0 ::1:53                  :::*                    LISTEN      2466/dnsmasq
tcp        0      0 fec0::5054:ff:fe12:3457:53 :::*                    LISTEN      2466/dnsmasq
udp        0      0 127.0.0.1:53            0.0.0.0:*               2466/dnsmasq
udp        0      0 192.168.2.15:53         0.0.0.0:*               2466/dnsmasq
```

在 Kali 的浏览器 里访问：<http://127.0.0.1:8080>，可以看到 OpenWrt 的 LuCI 登录页面：



4. ssh 连接

重新启动 OpenWrt 系统中的 SSH 服务：

```
/etc/init.d/dropbear restart
```

在 Kali 里新开一个终端窗口，输入：

```
ssh root@127.0.0.1 -p 1122
```

由于是第一次连接，中间过程需要输入 yes


```
File Actions Edit View Help
(parallels@kali-linux-2024-2)-[~]
$ ssh root@127.0.0.1 -p 1122
The authenticity of host '[127.0.0.1]:1122 ([127.0.0.1]:1122)' can't be established.
ED25519 key fingerprint is SHA256:bSKlW9J7NXWYB/H+Bld4ToASxSQ6e0dyYHGhJ5epsFY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[127.0.0.1]:1122' (ED25519) to the list of known hosts.

BusyBox v1.36.1 (2023-11-14 13:38:11 UTC) built-in shell (ash)

  _   _   _   _   _   _   _   _   _   _   _   _   _   _   _   _   _   _   _   _
 | |   | |   | |   | |   | |   | |   | |   | |   | |   | |   | |   | |   | | | | | | | | |
 | |_|  |_|  |_|  |_|  |_|  |_|  |_|  |_|  |_|  |_|  |_|  |_|  |_|  |_|  |_|  |_|
 |_|   |_|   |_|   |_|   |_|   |_|   |_|   |_|   |_|   |_|   |_|   |_|   |_|

  _   _   _   _   _   _   _   _   _   _   _   _   _   _   _   _   _   _   _
 | |   | |   | |   | |   | |   | |   | |   | |   | |   | |   | |   | |   | | | | | | | | |
 | |_|  |_|  |_|  |_|  |_|  |_|  |_|  |_|  |_|  |_|  |_|  |_|  |_|  |_|  |_|  |_|
 |_|   |_|   |_|   |_|   |_|   |_|   |_|   |_|   |_|   |_|   |_|   |_|   |_|

OpenWrt 23.05.2, r23630-842932a63d

==== WARNING! ====
There is no root password defined on this device!
Use the "passwd" command to set up a new password
in order to prevent unauthorized SSH logins.

root@OpenWrt:~#
```

由上图可以看出SSH 连接成功

四、实验总结

本次实验成功在 QEMU 上搭建了 ARMv8 架构的 OpenWrt 仿真环境，能够通过 U-Boot 正常引导系统并进入控制台，说明镜像与虚拟硬件配置匹配正确。在网络配置方面，通过 user 模式网络与端口转发实现了宿主机到虚拟机的访问：一方面可在浏览器通过 127.0.0.1:8080 打开 LuCI 管理界面，验证了 uhttpd Web 服务正常工作；另一方面可使用 ssh root@127.0.0.1 -p 1122 远程登录，验证了 Dropbear SSH 服务可用。

实验过程中加深了对 QEMU 关键参数（机器类型、CPU/内存、virtio 设备、网络与端口映射）的理解，也体会到在无图形界面（-nographic）条件下通过串口控制台进行系统管理的常见方式。总体而言，本实验完成了 OpenWrt 启动、Web 管理与远程登录三项核心验证，为后续在仿真环境中开展网络配置与服务部署实验打下了基础。